

Package ‘Rssofqrm’

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Title Robust Spatial Scalar-on-Function Quantile Regression Model
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Description Functions for implementing methods for robust spatial scalar-on-function quantile regression model.
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covid19_data	<i>COVID-19 Dataset</i>
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Description

A structured list object containing functional, scalar, and spatial components derived from archived COVID-19 data.

Usage

covid19_data

Format

A list with five components:

train A list containing confirmed cases and death numbers (mean) in the year 2021 for the train set.

test A list containing confirmed cases and death numbers (mean and upper quantiles) in the year 2022 for the test set.

latitude A numeric vector of latitude.

longitude A numeric vector of longitude.

weight_matrix A spatial weight matrix.

Details

The data were archived from the COVID19 R package and restructured for use in the robust functional spatial quantile regression framework implemented in this package.

Source

<https://cran.r-project.org/web/packages/COVID19/index.html>

data_generation

Data Generating Process for Spatial Scalar-on-Function Model

Description

Generates synthetic data from a spatial scalar-on-function regression model.

Usage

```
data_generation(n, j, rho, mean.e, sig.e, out.p = 0)
```

Arguments

n	Integer. Sample size.
j	Integer. Number of grid points.
rho	Numeric. Spatial autoregressive parameter.
mean.e	Numeric. Mean of error terms.
sig.e	Numeric. Standard deviation of error Terms.
out.p	Numeric. Proportion of outliers to introduce (default is 0).

Value

A list with the following elements:

y Response vector of length n.

x Matrix of functional predictor values.

w Spatial weight matrix of size n x n.

tcoefs True coefficient function used to generate the data.

out.index Indices of outlier observations, if any.

Examples

```
## Not run:
sim_data <- data_generation(n=250, j=101, rho=0.5, mean.e=1, sig.e=1, out.p = 0)
y <- sim_data$y
x <- sim_data$x
w <- sim_data$w
tcoef <- sim_data$tcoefs
out.index <- sim_data$out.index

## End(Not run)
```

predict_rssofqr	<i>Predict Method for Robust Functional Quantile Regression with Spatial Effects</i>
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Description

Makes predictions for new observations using a fitted robust spatial functional quantile regression model.

Usage

```
predict_rssofqr(object, xnew, wnew)
```

Arguments

object	A fitted model object returned by rssofqr .
xnew	Numeric matrix of size $n \times p$. New functional predictor data (on the same grid used in training).
wnew	Numeric matrix of size $n \times n$. Spatial weight matrix for the new observations.

Value

A numeric vector of predicted values of length n .

Examples

```
## Not run:
sim_data <- data_generation(n=250, j=101, rho=0.5, mean.e=1, sig.e=1, out.p = 0)
y <- sim_data$y
x <- sim_data$x
w <- sim_data$w
fit_kim <- rssofqr(y=y, x=x, w=w, tau=tau, model = "KM", CV="TRUE")
fit_ch <- rssofqr(y=y, x=x, w=w, tau=tau, model = "CH", CV="TRUE")
sim_test <- data_generation(n=250, j=101, rho=0.5, mean.e=1, sig.e=1, out.p = 0)
y_test <- sim_test$y
x_test <- sim_test$x
w_test <- sim_test$w
predict_kim <- predict_rssofqr(object = fit_kim, xnew = x_test, wnew = w_test)
predict_ch <- predict_rssofqr(object = fit_ch, xnew = x_test, wnew = w_test)

## End(Not run)
```

 rssofqr

Robust Functional Quantile Regression Model with Spatial Autocorrelation

Description

Fits a spatial scalar-on-function quantile regression model using either the Kim and Muller (2004) or Chernozhukov and Hansen (2006) method.

Usage

```
rssofqr(
  y,
  x,
  w,
  tau,
  tau_comp = NULL,
  gp = NULL,
  nbasis = NULL,
  h = NULL,
  model = c("KM", "CH"),
  CV = c("TRUE", "FALSE")
)
```

Arguments

<code>y</code>	Numeric vector of length n . Response variable.
<code>x</code>	Numeric matrix of size $n \times p$. Functional predictor observed on a common grid.
<code>w</code>	Numeric matrix of size $n \times n$. Spatial weight matrix.
<code>tau</code>	Numeric. Quantile level to estimate (between 0 and 1).
<code>tau_comp</code>	Numeric. Quantile level to estimate for extracted components (between 0 and 1).
<code>gp</code>	Numeric vector of length p . Grid of argument values for the functional predictor. If NULL, defaults to a regular grid between 0 and 1.
<code>nbasis</code>	Integer. Number of basis functions to use in RFPQR. If NULL, defaults to $\min(10, p / 4)$.
<code>h</code>	Integer. Number of extracted components. If NULL, it is optimally determined when CV is True.
<code>model</code>	Character string. Estimation method to use. One of "KM" (Kim et al.) or "CH" (Chernozhukov et al.).
<code>CV</code>	Character string. Determine the number of components to extract based on the BIC. One of "TRUE" or "FALSE".

Value

A list containing:

- b** Estimated coefficient vector in RFPQR space.
- b0** Estimated intercept.

bhat Estimated coefficient function in the original function space.

rho Estimated spatial autoregressive parameter.

opt.h Number of extracted optimal components.

opt.nbasis Number of basis functions.

fitted.values Fitted values from the model.

residuals Residuals from the model.

tau Quantile level used.

details Output from getRFPQR, including RFPQR scores and basis functions.

Examples

```
## Not run:
sim_data <- data_generation(n=250, j=101, rho=0.5, mean.e=1, sig.e=1, out.p = 0)
y <- sim_data$y
x <- sim_data$x
w <- sim_data$w
fit_kim <- rssofqr(y=y, x=x, w=w, tau=tau, model = "KM", CV="TRUE")
fit_ch <- rssofqr(y=y, x=x, w=w, tau=tau, model = "CH", CV="TRUE")

## End(Not run)
```

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